

NINAD NAIK

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EDUCATION

Bachelor of Engineering in Information Technology

University of Mumbai, Thadomal Shahani Engineering College (TSEC), Mumbai, India

2021 - 2025

CGPA: 8.54 / 10

WORK EXPERIENCE

Associate Backend Developer, Commotion (acq. by Tata Communications) (gocommotion.com)

January 2025 - Present

- Promoted from Backend Developer Intern (January 2025 - June 2025)
- Engineered **low-latency backend architectures** using NestJS and FastAPI, utilizing targeted database indexes and query optimization to handle **high-concurrency enterprise workloads**.
- Developed **MCP frameworks and A2A orchestration patterns** for real-time contextual voice and text AI agents.
- Built robust **AI integrations** with platforms like Salesforce, ServiceNow, and SAP, unifying multi-system workflows.
- Enforced security using OAuth and JWT auth schemes, ensuring platform integrity via SAST and VAPT security analysis.
- Developed **low-latency Voice AI** solutions in collaboration with **multiple Tata companies and MNCs** to streamline automated operations.

Full Stack Developer Intern, Arxena Inc. (arxena.com)

June 2024 - July 2024

- Improved efficiency of the automated recruitment process by developing a **GPT-powered WhatsApp chatbot** in NestJS, **WebSocket API, GraphQL and React.js**, streamlining communication with candidates.
- Optimized build process of the apps running on AWS EC2 using AMI, reducing costs by upto 75% (detailed in [Gist](#)).

OPEN SOURCE MENTORSHIP

The Linux Foundation - LFX - Linux Kernel Mentorship Program mentee, Spring 2026 ([link](#))

March 2026 - August 2026

- Graduated with **11 patches contributed** to Linux Kernel mainline; published a [blog](#) sharing my experience.
- Converted text-based **Devicetree bindings** to JSON Schema; fixed documentation links, typos, and driver error handling.
- Worked with **kernel development workflow, patch submission, and review cycles** and used tools like **cscope, Coccinelle, and smatch** for static analysis. Selected for the mentorship program from **693 applicants**.

PROJECTS

Redspot, FastAPI, React.js, Clickhouse, Kafka, PostgreSQL, Docker | ([GitHub](#))

June 2026

- Built a **real-time website click analytics platform** with a FastAPI ingestion server, Kafka message queue, and ClickHouse storage pipeline that captures user interactions via a lightweight JavaScript tracking script.
- Developed a React dashboard that renders target websites in an iframe and overlays a **color-coded heatmap visualization of click density** using canvas, powered by live data queried from ClickHouse

SpeakSure, Python, Flask, Tensorflow, React.js, MongoDB | ([GitHub](#))

October 2025

- Developed an **Automated AI Behavioural Interview system** that quantifies **confidence level** in candidate's speech and provides AI insights to the hiring manager. Presented **research paper** at ICCIS 25 based on new models and findings.
- Trained **multilayer perceptron (MLP)** and **Fuzzy Hypersphere Neural Network** models on a custom made dataset with **751** audio samples of 30 seconds each by extracting features with Wav2Vec2 and achieved **96%** relaxed testing accuracy.

Shortomega, React.js, Next.js, Node.js, NestJS, TypeScript, Redis, Docker | ([Website](#), [GitHub](#))

November 2024

- Developed a **Full Stack URL Shortening App** that generates short URLs for long URLs, with a focus on system design.
- Published a [technical blog](#) describing the methods of using **Redis as a Primary Database**.
- Implemented **Dashboard, Analytics data aggregation and Rate Limiting** features, and **Dockerized** the application.

SKILLS

Languages: TypeScript, JavaScript, Python, Go, C, C++, Java

Backend & APIs: NestJS, FastAPI, Node.js, GraphQL, gRPC

AI & Security: MCP, A2A, OAuth2.0, JWT

Frontend: React.js, Tailwind CSS, HTML, CSS

Data & Messaging: PostgreSQL, MongoDB, Redis, Kafka

Platforms & Tools: Linux, Docker, Kubernetes, AWS

RESEARCH AND PUBLICATION

Evaluating Human Speech Confidence with Deep Learning Techniques, Ninad Naik, et al.

September 2025

- Validated **Multi Layer Perceptron, Fuzzy Hypersphere Neural Network** and **Logistic Regression** classifiers over **MFCC** and **Wav2Vec2** features on a custom **751-recording** dataset using **5-fold stratified cross-validation**, reaching **96% relaxed accuracy** across 5 confidence levels.
- Presented at **ICCIS 2025**, BITS Pilani, Goa; accepted for publication in SCOPUS-indexed **Springer LNNS**. ([Certificate](#))

ADDITIONAL INFORMATION

- Open Source Contribution to the [Linux Kernel](#), [n8n.io](#) and [CircuitVerse](#).
- First Runner Up (CSS Battle 2022 & Javagenix 2022), 2nd Runner Up (PyExpo 2023), Top 10 (ACE Hacks 2024)
- Maintaining a **tech blog** to share experiences, learnings and results on tech.ninadnaik.me